MACSE603 — LABORATORY RECORD

Assessment 2

Visual Analytics Lab

DAX Calculations, Advanced Filter Context and Report Publishing

Power BI Desktop · DAX · Power BI Service

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M.Tech — Computer Science and Engineering

Power BI Desktop · SQL Server · Power Query (M) · DAX · Python

Name: Vedant Nimkar | **Registration No.:** 26MML0045 | **Subject:** Visual Analytics Lab (MACSE603)
Assessment 2 covers: calculated tables, columns and measures in DAX; advanced DAX with `CALCULATE()` and Time Intelligence; report design and publishing.

Aim

4. Create calculated tables, calculated columns and measures using Data Analysis Expressions, to answer questions such as the top sold product and the average sales of a product category.
5. Use the `CALCULATE()` function to manipulate filter context, and use Time Intelligence functions on the sales data — for example total sales revenue for each product category and the year-over-year growth trend of a product category.
6. Design a report with the sales data and publish it with basic visualizations.

Prerequisite

Assessment 1 must be complete: the cleansed `FactSales` table of 5,013 rows loaded, the star schema built with four active relationships, and `DimDate` marked as a date table. Time Intelligence functions in Step 5 return silently incorrect results if the date table is not marked.

Step 4 — Calculated Tables, Columns and Measures

4.1 Calculated columns

Created via **Table tools → New column**. A calculated column is evaluated row by row at refresh time and stored in the model.

```
Net Unit Price = DIVIDE ( FactSales[SalesAmount], FactSales[OrderQuantity] )

Line Margin % =
DIVIDE ( FactSales[SalesAmount] - FactSales[TotalCost],
         FactSales[SalesAmount] )

Order Size Band =
SWITCH ( TRUE (),
    FactSales[OrderQuantity] <= 2, "1-2 units",
    FactSales[OrderQuantity] <= 4, "3-4 units",
    FactSales[OrderQuantity] <= 6, "5-6 units",
    "7+ units" )

Product Category = RELATED ( DimProduct[Category] )

Price Tier =
SWITCH ( TRUE (),
    DimProduct[ListPrice] < 100, "Budget",
    DimProduct[ListPrice] < 500, "Mid-range",
    "Premium" )

Financial Year =
"FY " & IF ( DimDate[MonthNo] >= 4, DimDate[Year], DimDate[Year] - 1 )
      & "-" & RIGHT ( IF ( DimDate[MonthNo] >= 4,
                           DimDate[Year] + 1, DimDate[Year] ), 2 )
```

4.2 Core measures

Created via **Table tools → New measure**. A measure is evaluated at query time against the current filter context and is not stored.

```
Total Sales           = SUM ( FactSales[SalesAmount] )
Total Quantity        = SUM ( FactSales[OrderQuantity] )
Total Cost            = SUM ( FactSales[TotalCost] )
Total Profit          = [Total Sales] - [Total Cost]
Profit Margin %       = DIVIDE ( [Total Profit], [Total Sales] )
Order Count           = DISTINCTCOUNT ( FactSales[SalesOrderNumber] )
Average Order Value   = DIVIDE ( [Total Sales], [Order Count] )
Average Line Sales    = AVERAGE ( FactSales[SalesAmount] )
Average Selling Price = DIVIDE ( [Total Sales], [Total Quantity] )
Customer Count        = DISTINCTCOUNT ( FactSales[CustomerKey] )
```

Observation — unfiltered model

Measure	Value
Total Sales	8,955,541.25
Total Quantity	22,448
Total Cost	4,123,542.00
Total Profit	4,831,999.25
Profit Margin %	53.96%
Order Count	5,008
Average Order Value	1,788.25

4.3 Question 1 — top sold product

```
Top Product by Units =
VAR TopProduct =
    TOPN ( 1, ALL ( DimProduct[ProductName] ),
           CALCULATE ( [Total Quantity] ), DESC )
RETURN CONCATENATEX ( TopProduct, DimProduct[ProductName], ", " )

Top Product Units =
MAXX ( ALL ( DimProduct[ProductName] ), CALCULATE ( [Total Quantity] ) )

Top Product by Revenue =
VAR TopProduct =
    TOPN ( 1, ALL ( DimProduct[ProductName] ),
           CALCULATE ( [Total Sales] ), DESC )
RETURN CONCATENATEX ( TopProduct, DimProduct[ProductName], ", " )

Product Rank by Units =
RANKX ( ALL ( DimProduct[ProductName] ), [Total Quantity], , DESC, DENSE )
```

`ALL()` is required inside `TOPN` so that the ranking is evaluated across every product rather than only the product on the current row.

Observation

Product	Units sold	Revenue
GlideMouse Wireless	**3,204**	141,847.22
TactKey Mechanical	2,501	249,113.16
GamingBook X17	1,344	**2,534,432.17**
Zenith A12	1,219	383,030.70
EduBook 13	1,210	746,661.45

The two rankings disagree, which is the analytically interesting result. `GlideMouse Wireless` leads on volume at 3,204 units but contributes only 141,847.22 in revenue, whereas `GamingBook X17` sells 58% fewer units yet generates almost eighteen times the revenue. A dashboard that reported only unit volume would misdirect attention entirely.

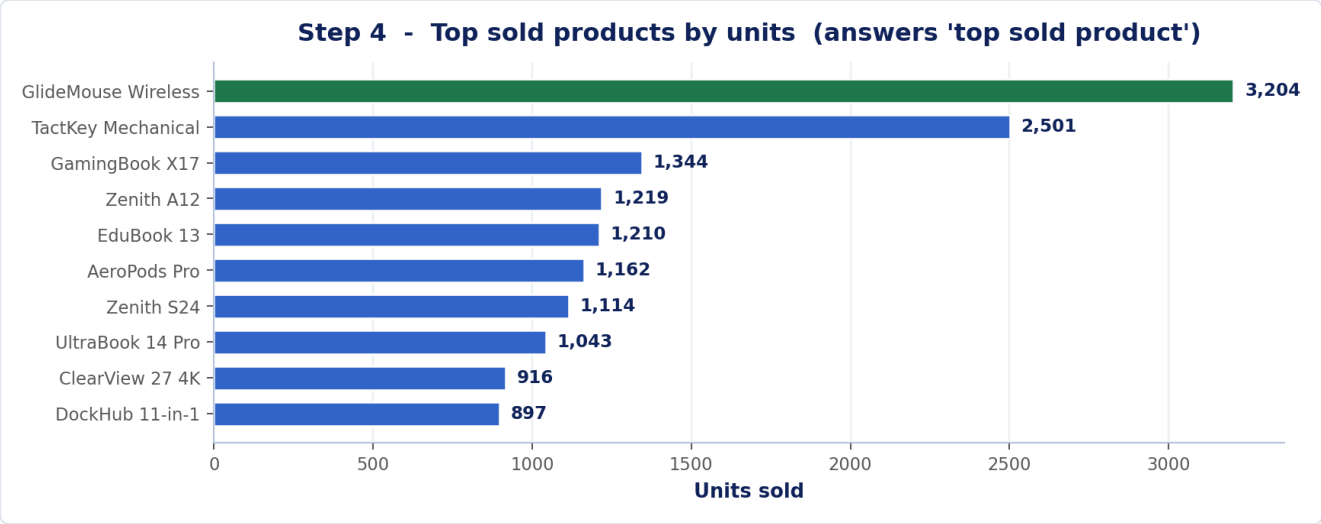


Figure 1. Top ten products by units sold, with the leading product highlighted.

4.4 Question 2 — average sales of product category

```
Avg Sales per Line = AVERAGE ( FactSales[SalesAmount] )

Avg Sales per Order =
DIVIDE ( [Total Sales], DISTINCTCOUNT ( FactSales[SalesOrderNumber] ) )

Avg Sales per Category =
AVERAGEX ( VALUES ( DimProduct[Category] ), [Total Sales] )

Category Share % =
DIVIDE ( [Total Sales], CALCULATE ( [Total Sales], ALL ( DimProduct ) ) )
```

Observation

Category	Total Sales	Orders	Units	Avg Sales per Order	Share %
Electronics	7,081,992.08	2,172	9,758	3,260.59	79.08%
Furniture	861,784.14	519	2,422	1,660.47	9.62%
Accessories	786,667.35	1,996	8,818	394.12	8.78%
Software	225,097.68	321	1,450	701.24	2.51%

`Avg Sales per Category` across all four categories = **2,238,885.31**.

Accessories ranks second on order count (1,996, almost matching Electronics at 2,172) yet last but one on total sales, because its average order value is 394.12 against Electronics at 3,260.59 — roughly eight times lower. High transaction volume is therefore not a proxy for revenue contribution.

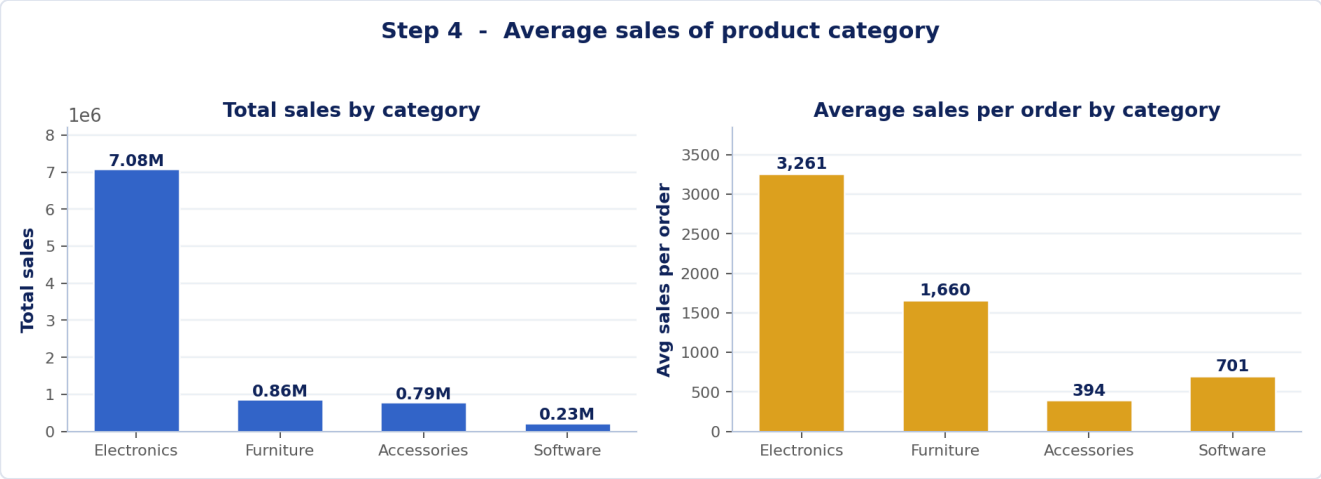


Figure 2. Total sales and average sales per order by category, showing that order count and order value rank the categories differently.

4.5 Calculated tables

Created via **Modeling → New table**. A calculated table is materialised at refresh.

```
Product Category Summary =
SUMMARIZE ( FactSales, DimProduct[Category],
    "Total Sales",         [Total Sales],
    "Total Quantity",      [Total Quantity],
    "Order Count",         [Order Count],
    "Avg Sales per Order", [Avg Sales per Order],
    "Total Profit",        [Total Profit],
    "Profit Margin %",     [Profit Margin %] )

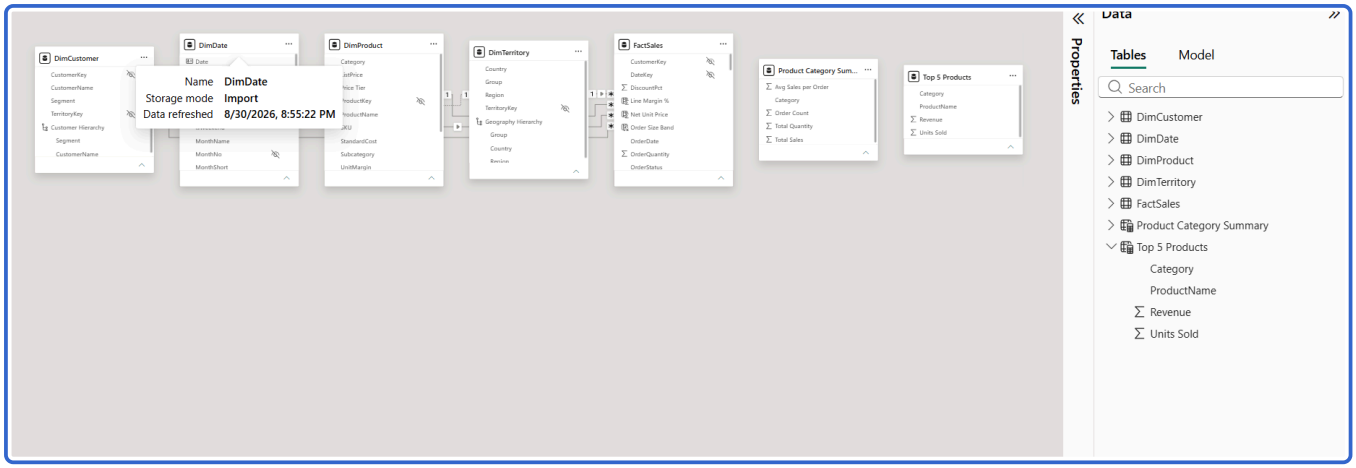
Top 5 Products =
TOPN ( 5,
    SUMMARIZE ( FactSales, DimProduct[ProductName], DimProduct[Category],
        "Units Sold", [Total Quantity], "Revenue", [Total Sales] ),
    [Units Sold], DESC )

Date Table DAX =
ADDCOLUMNS ( CALENDAR ( DATE(2023,1,1), DATE(2025,12,31) ),
    "Year",      YEAR ( [Date] ),
    "MonthNo",   MONTH ( [Date] ),
    "MonthName", FORMAT ( [Date], "MMMM" ),
    "Quarter",   "Q" & QUARTER ( [Date] ),
    "YearMonth", FORMAT ( [Date], "YYYY-MM" ) )

Category Year Performance =
SUMMARIZE ( FactSales, DimProduct[Category], DimDate[Year],
            "Sales", [Total Sales] )

Discount Scenario = GENERATESERIES ( 0, 0.25, 0.05 )
```

`Product Category Summary` returns exactly four rows matching the table in section 4.4.



Screenshot 1. Product Category Summary calculated table in Data view

Step 5 — Advanced DAX: CALCULATE and Time Intelligence

5.1 Manipulating filter context with CALCULATE()

`CALCULATE` evaluates an expression under a modified filter context. It is the only DAX function that can change filter context.

Technique	Function	Effect on context
Add a filter	`Category = "Electronics"`	narrows the context
Remove all filters	`ALL(DimProduct)`	grand-total denominator
Keep one filter	`ALLEXCEPT(DimProduct, Category)`	share within own category
Respect slicers	`ALLSELECTED(Category)`	share of user selection
Intersect, not replace	`KEEPFILTERS(Year = 2025)`	preserves an existing Year slicer
Switch relationship	`USERELATIONSHIP(...)`	activates the inactive path from Assessment 1

```
Electronics Sales =
CALCULATE ( [Total Sales], DimProduct[Category] = "Electronics" )

Pct of Total Sales =
DIVIDE ( [Total Sales],
         CALCULATE ( [Total Sales], ALL ( DimProduct ),
                                    ALL ( DimTerritory ) ) )

Pct of Category Sales =
DIVIDE ( [Total Sales],
         CALCULATE ( [Total Sales],
                     ALLEXCEPT ( DimProduct, DimProduct[Category] ) ) )

Sales 2025 Override    = CALCULATE ( [Total Sales], DimDate[Year] = 2025 )
Sales 2025 KeepFilters =
CALCULATE ( [Total Sales], KEEPFILTERS ( DimDate[Year] = 2025 ) )

Laptops or Mobiles Sales =
CALCULATE ( [Total Sales],
    FILTER ( ALL ( DimProduct[Subcategory] ),
             DimProduct[Subcategory] = "Laptops"
          || DimProduct[Subcategory] = "Mobiles" ) )

Sales by Customer Territory =
CALCULATE ( [Total Sales],
    USERELATIONSHIP ( DimCustomer[TerritoryKey],
                      DimTerritory[TerritoryKey] ) )
```

`Sales 2025 Override` and `Sales 2025 KeepFilters` differ materially: a direct predicate **replaces** any existing filter on that column, so with a Year slicer set to 2023 the override still returns 2025. `KEEPFILTERS` **intersects** instead, returning blank because 2023 and 2025 do not overlap.

5.2 Total sales revenue for each product category

```
Category Revenue =
CALCULATE ( [Total Sales], ALLEXCEPT ( DimProduct, DimProduct[Category] ) )

Category Revenue Table =
SUMMARIZE ( FactSales, DimProduct[Category], "Revenue", [Total Sales] )
```

Category	2023	2024	2025	Total
Electronics	2,229,091.18	2,422,281.89	2,430,619.01	7,081,992.08
Furniture	215,300.29	285,115.53	361,368.32	861,784.14
Accessories	214,422.23	264,027.21	308,217.91	786,667.35
Software	61,936.51	61,233.78	101,927.39	225,097.68
Total	**2,720,750.21**	**3,032,658.41**	**3,202,132.63**	**8,955,541.25**

5.3 Time Intelligence

```
Sales YTD = TOTALYTD ( [Total Sales], DimDate[Date] )
Sales QTD = TOTALQTD ( [Total Sales], DimDate[Date] )
Sales MTD = TOTALMTD ( [Total Sales], DimDate[Date] )

-- Indian financial year, April start
Sales FYTD = TOTALYTD ( [Total Sales], DimDate[Date], "31/03" )

Sales PY = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( DimDate[Date] ) )
Sales PM = CALCULATE ( [Total Sales], DATEADD ( DimDate[Date], -1, MONTH ) )

Sales Rolling 3M =
CALCULATE ( [Total Sales],
    DATESINPERIOD ( DimDate[Date], MAX ( DimDate[Date] ), -3, MONTH ) )

Sales Cumulative =
CALCULATE ( [Total Sales],
    FILTER ( ALLSELECTED ( DimDate[Date] ),
             DimDate[Date] <= MAX ( DimDate[Date] ) ) )

Sales 3M Moving Avg =
AVERAGEX (
    DATESINPERIOD ( DimDate[Date], MAX ( DimDate[Date] ), -3, MONTH ),
    CALCULATE ( [Total Sales] ) )
```

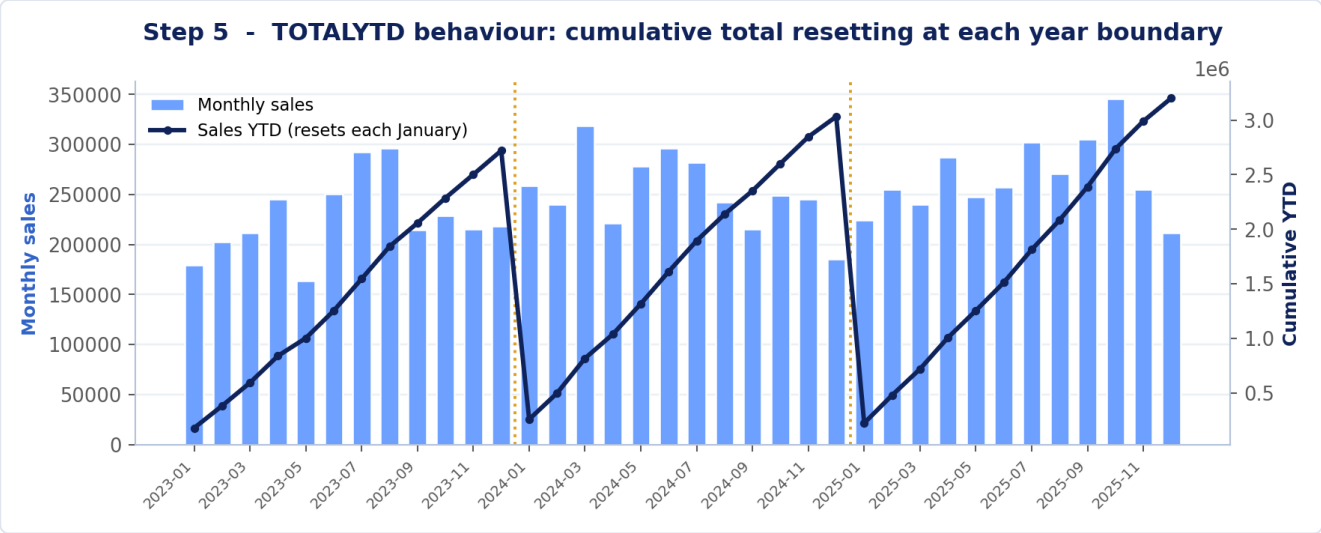


Figure 3. `TOTALYTD` behaviour. The cumulative line resets at each year boundary, confirming `DimDate` is correctly marked as a date table.

5.4 Year-over-year growth trend of a product category

```
YoY Growth =
VAR Cur = [Total Sales]
VAR Prev = [Sales PY]
RETURN IF ( NOT ISBLANK ( Prev ), Cur - Prev )

YoY Growth % = DIVIDE ( [Total Sales] - [Sales PY], [Sales PY] )

YoY Trend =
VAR G = [YoY Growth %]
RETURN SWITCH ( TRUE (),
    ISBLANK ( G ), "No prior year",
    G > 0.10,      "Strong growth",
    G > 0,         "Growing",
    G > -0.10,     "Slight decline",
    "Sharp decline" )

Electronics YoY % =
VAR Cur  = CALCULATE ( [Total Sales], DimProduct[Category] = "Electronics" )
VAR Prev = CALCULATE ( [Total Sales], DimProduct[Category] = "Electronics",
                       SAMEPERIODLASTYEAR ( DimDate[Date] ) )
RETURN DIVIDE ( Cur - Prev, Prev )
```

Observation

Category	2024 vs 2023	2025 vs 2024	Trend
Accessories	+23.13%	+16.74%	Strong growth
Electronics	+8.67%	+0.34%	Growing, now plateaued
Furniture	+32.43%	+26.74%	Strong growth
Software	−1.13%	+66.45%	Recovered sharply
Overall	**+11.46%**	**+5.59%**	Growing

The headline figures conceal the important detail. Overall growth halved from +11.46% to +5.59%, and the cause is visible only at category level: Electronics, which is 79% of revenue, effectively stopped growing at +0.34%. Furniture and

Accessories continued to grow strongly but are too small to compensate. Software reversed from −1.13% to +66.45%, though from a low base of 2.51% of revenue. This is precisely the divergence that Time Intelligence measures exist to surface.

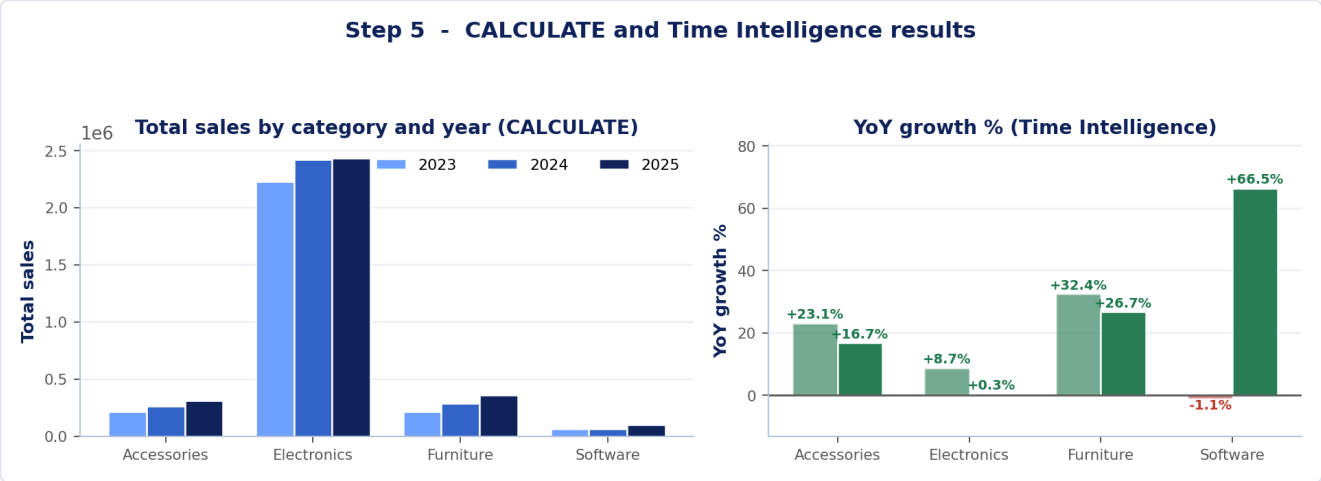


Figure 4. Category revenue by year produced with `CALCULATE`, and year-over-year growth percentage produced with `SAMEPERIODLASTYEAR`.

Step 6 — Designing and Publishing the Report

6.1 Page 1 — Sales Overview

Element	Visual type	Configuration
KPI row	Cards	Total Sales, Total Quantity, Total Profit, Profit Margin %, YoY Growth %
Sales trend	Line chart	Axis `Date Hierarchy`; values `[Total Sales]`, `[Sales PY]`
Category performance	Stacked column	Axis `Category`, legend `Year`
Top 10 products	Bar chart	Top-N filter 10 by `[Total Quantity]`
Regional split	Map or donut	Location `Region`
Category detail	Table	`Category`, `Subcategory`, `[Total Sales]`, `[Avg Sales per Order]`, `[Category Share %]`
Slicers	Year, Category, Region, Segment, OrderStatus	left panel

6.2 Page 2 — Category and Time Analysis

Element	Visual type	Configuration
YoY matrix	Matrix	Rows `Category`, columns `Year`, values `[Total Sales]`, `[Sales PY]`, `[YoY Growth %]`
Cumulative sales	Area chart	Axis `YearMonth`, value `[Sales Cumulative]`
Rolling average	Line chart	`[Total Sales]` vs `[Sales 3M Moving Avg]`
Monthly YTD	Line and clustered column	column `[Total Sales]`, line `[Sales YTD]`
Growth by subcategory	Bar chart	Axis `Subcategory`, value `[YoY Growth %]`

Conditional formatting is applied to `[YoY Growth %]` on the matrix: green background above zero, red below.

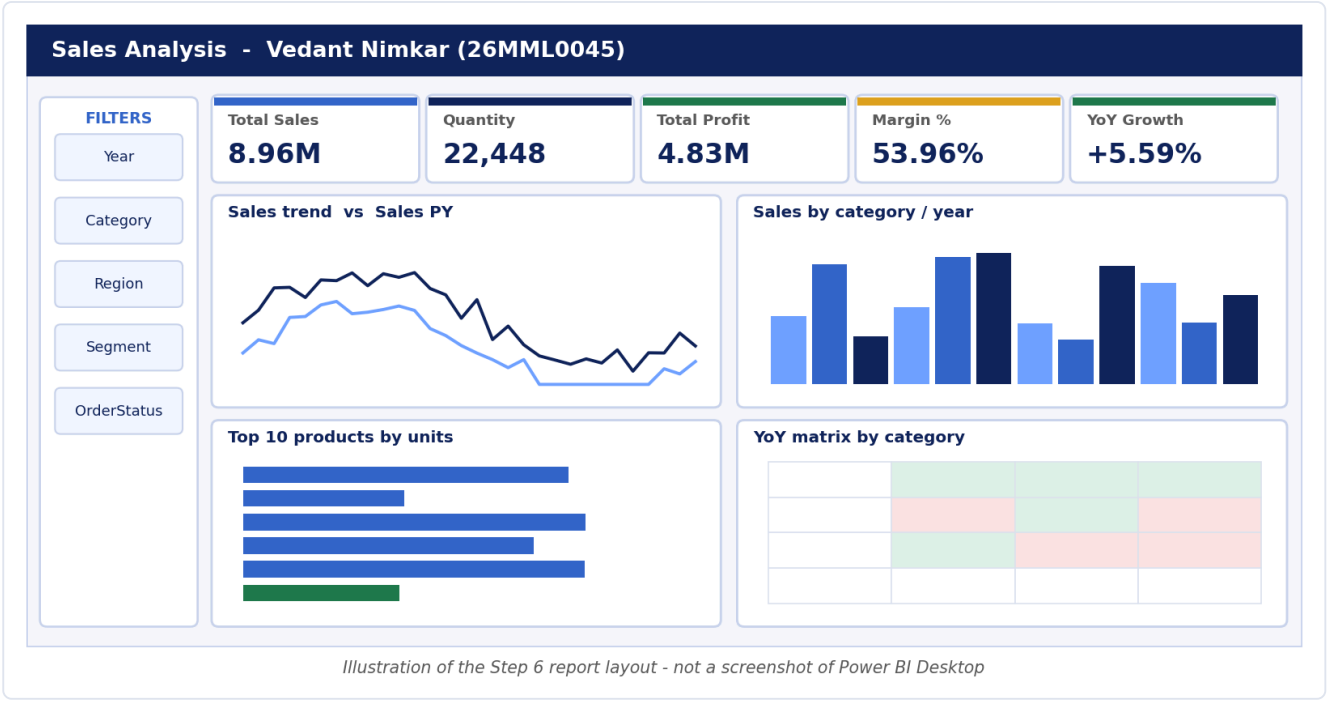
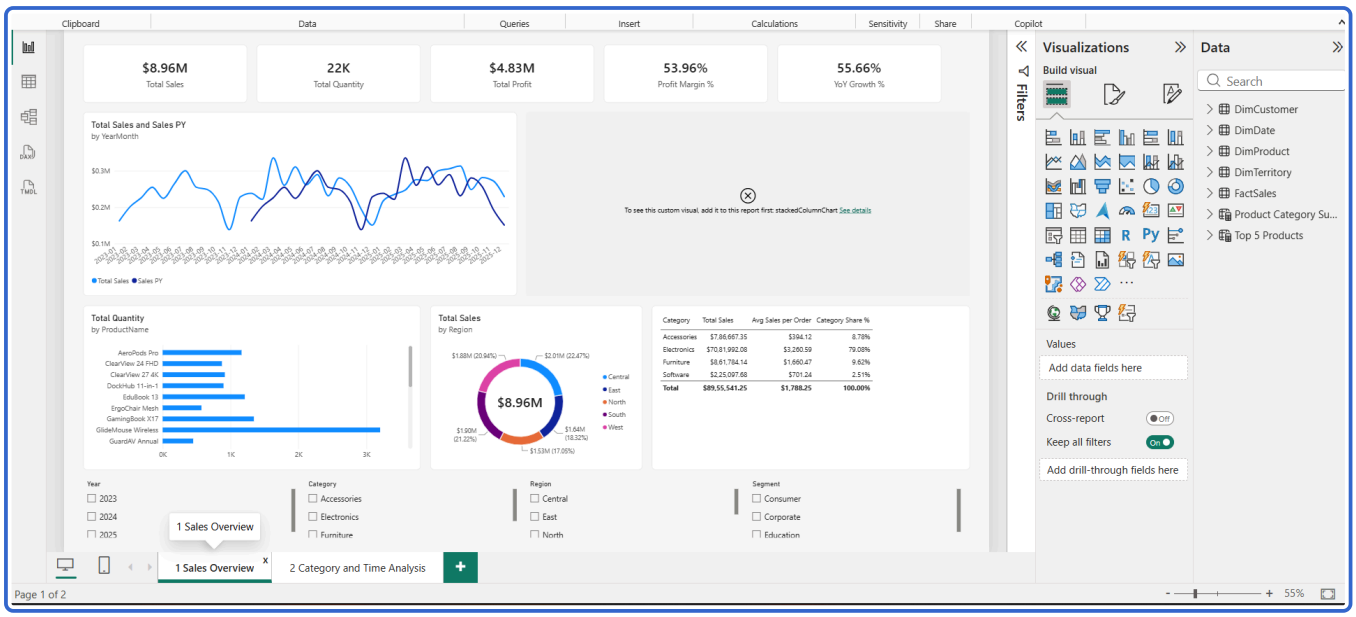
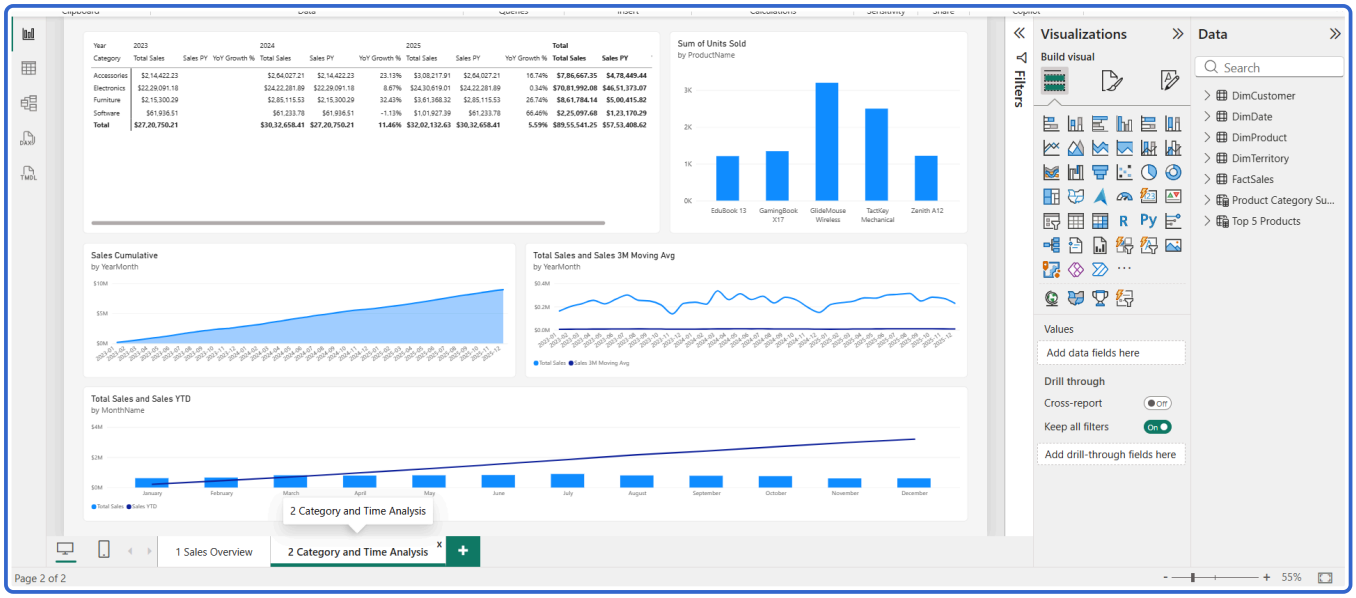


Illustration of the Step 6 report layout - not a screenshot of Power BI Desktop

Figure 5. Intended layout of the Page 1 report canvas. This is an illustration of the design, not a Power BI capture.



Screenshot 2. Report Page 1 — Sales Overview in Power BI



Screenshot 3. Report Page 2 — Category and Time Analysis in Power BI

6.3 Formatting applied

Setting	Value
Canvas	16:9, background `#F7F9FC`
Accent colour	`#3366CC` throughout
Page title	`Sales Analysis – Vedant Nimkar (26MML0045)`
Drill-down	enabled via `Product Hierarchy` on category visuals
Interactions	slicers set to filter rather than highlight
Sort order	`[Total Sales]` descending on all category visuals

6.4 Publishing procedure

1. **File → Save As →** `VisualAnalytics_Sales_26MML0045.pbix`
2. **Home → Publish**
3. Sign in with the institutional Microsoft 365 account
4. Select destination workspace → **My workspace**
5. On the *Success!* confirmation, follow the link to open the report in the browser

Post-publish configuration

Task	Location
Scheduled refresh	Workspace → Dataset → Settings → Scheduled refresh
Gateway connection	required only for the SQL Server source
Pin visuals to dashboard	hover a visual → Pin icon
Export	File → Export to PDF

Result

All three objectives of Assessment 2 were completed.

Step 4. Six calculated columns, ten core measures and five calculated tables were authored in DAX. `TOPN` with `ALL` identified `GlideMouse Wireless` as the top product by units at 3,204, while ranking by revenue returned `GamingBook X17` at 2,534,432.17 — demonstrating that volume and value produce different leaders. Category averages showed Accessories generating an average order value of 394.12 against Electronics at 3,260.59 despite comparable order counts.

Step 5. `CALCULATE` was used with `ALL`, `ALLEXCEPT`, `ALLSELECTED`, `KEEPFILTERS`, `FILTER` and `USERELATIONSHIP` to add, remove, retain and intersect filters and to activate the inactive relationship from Assessment 1. Time Intelligence functions produced year-to-date, prior-year, rolling three-month, cumulative and moving-average measures. Category year-over-year growth ranged from −1.13% to +66.45% against an overall +5.59%, revealing that Electronics had plateaued at +0.34% while contributing 79% of revenue.

Step 6. A two-page report was designed with KPI cards, trend, category, ranking and regional visuals, drill-down hierarchies, five slicers and a conditionally formatted year-over-year matrix, then published to the Power BI Service.

Every measure value was cross-checked against an independent pandas computation over the same source data, recorded in `expected_results/EXPECTED_RESULTS.txt`.

Deliverables

Path	Contents
`code/Step4_DAX_Calculated.txt`	calculated columns, measures, tables
`code/Step5_DAX_Advanced.txt`	CALCULATE and Time Intelligence
`code/Step6_Report_Design.md`	report layout and publish steps
`expected_results/`	verification values and derived tables
`figures/`	figures reproduced in this record
`screenshots/`	Power BI Desktop captures

Appendix — Complete Applied DAX

The listing below was extracted programmatically from the applied semantic model (`Visual1.SemanticModel/definition/tables/*.tmdl`). It is therefore the exact code running in the report, not a separately maintained transcription.

Object type	Count
Tables	7
Calculated columns	6
Measures	51
Calculated tables	2

Calculated columns

DimDate

```
Financial Year =
    "FY " & IF ( DimDate[MonthNo] >= 4, DimDate[Year],
                    DimDate[Year] - 1 )
          & "-" & RIGHT ( IF ( DimDate[MonthNo] >= 4,
                    DimDate[Year] + 1, DimDate[Year] ), 2 )
```

DimProduct

```
Price Tier =
    SWITCH ( TRUE (),
        DimProduct[ListPrice] < 100, "Budget",
        DimProduct[ListPrice] < 500, "Mid-range",
        "Premium" )
```

FactSales

```
Net Unit Price =
    DIVIDE ( FactSales[SalesAmount], FactSales[OrderQuantity] )

Line Margin % =
    DIVIDE ( FactSales[SalesAmount] - FactSales[TotalCost],
             FactSales[SalesAmount] )

Order Size Band =
    SWITCH ( TRUE (),
        FactSales[OrderQuantity] <= 2, "1-2 units",
        FactSales[OrderQuantity] <= 4, "3-4 units",
        FactSales[OrderQuantity] <= 6, "5-6 units",
        "7+ units" )

Product Category =
    RELATED ( DimProduct[Category] )
```

Measures

01 Sales

```
Total Sales =
    SUM ( FactSales[SalesAmount] )

Total Quantity =
    SUM ( FactSales[OrderQuantity] )

Order Count =
    DISTINCTCOUNT ( FactSales[SalesOrderNumber] )

Average Order Value =
    DIVIDE ( [Total Sales], [Order Count] )

Average Line Sales =
    AVERAGE ( FactSales[SalesAmount] )

Average Selling Price =
    DIVIDE ( [Total Sales], [Total Quantity] )

Customer Count =
    DISTINCTCOUNT ( FactSales[CustomerKey] )

Product Count =
    DISTINCTCOUNT ( FactSales[ProductKey] )
```

02 Profit

```
Total Cost =
    SUM ( FactSales[TotalCost] )

Total Profit =
    [Total Sales] - [Total Cost]

Profit Margin % =
    DIVIDE ( [Total Profit], [Total Sales] )
```

03 Ranking

```
Top Product by Units =
    VAR TopProduct =
        TOPN ( 1, ALL ( DimProduct[ProductName] ),
               CALCULATE ( [Total Quantity] ), DESC )
    RETURN
        CONCATENATEX ( TopProduct, DimProduct[ProductName], ", " )

Top Product Units =
    MAXX ( ALL ( DimProduct[ProductName] ),
           CALCULATE ( [Total Quantity] ) )

Top Product by Revenue =
    VAR TopProduct =
        TOPN ( 1, ALL ( DimProduct[ProductName] ),
               CALCULATE ( [Total Sales] ), DESC )
    RETURN
        CONCATENATEX ( TopProduct, DimProduct[ProductName], ", " )

Product Rank by Units =
    RANKX ( ALL ( DimProduct[ProductName] ), [Total Quantity], ,
            DESC, DENSE )
```

04 Category

```
Avg Sales per Line =
    AVERAGE ( FactSales[SalesAmount] )

Avg Sales per Order =
    DIVIDE ( [Total Sales],
             DISTINCTCOUNT ( FactSales[SalesOrderNumber] ) )

Avg Sales per Category =
    AVERAGEX ( VALUES ( DimProduct[Category] ), [Total Sales] )

Category Share % =
    DIVIDE ( [Total Sales],
             CALCULATE ( [Total Sales], ALL ( DimProduct ) ) )
```

05 CALCULATE

```
Electronics Sales =
    CALCULATE ( [Total Sales], DimProduct[Category] = "Electronics" )

Shipped Sales =
    CALCULATE ( [Total Sales], FactSales[OrderStatus] = "Shipped" )

Discounted Sales =
    CALCULATE ( [Total Sales], FactSales[DiscountPct] > 0 )

Sales All Products =
    CALCULATE ( [Total Sales], ALL ( DimProduct ) )

Pct of Total Sales =
    DIVIDE ( [Total Sales],
             CALCULATE ( [Total Sales], ALL ( DimProduct ),
                                        ALL ( DimTerritory ) ) )

Category Revenue =
    CALCULATE ( [Total Sales],
                ALLEXCEPT ( DimProduct, DimProduct[Category] ) )

Pct of Category Sales =
    DIVIDE ( [Total Sales],
             CALCULATE ( [Total Sales],
                 ALLEXCEPT ( DimProduct, DimProduct[Category] ) ) )

Pct of Selected Sales =
    DIVIDE ( [Total Sales],
             CALCULATE ( [Total Sales],
                 ALLSELECTED ( DimProduct[Category] ) ) )

Sales 2025 Override =
    CALCULATE ( [Total Sales], DimDate[Year] = 2025 )

Sales 2025 KeepFilters =
    CALCULATE ( [Total Sales],
                KEEPFILTERS ( DimDate[Year] = 2025 ) )

Laptops or Mobiles Sales =
    CALCULATE ( [Total Sales],
        FILTER ( ALL ( DimProduct[Subcategory] ),
                  DimProduct[Subcategory] = "Laptops"
              || DimProduct[Subcategory] = "Mobiles" ) )

Sales by Customer Territory =
    CALCULATE ( [Total Sales],
        USERELATIONSHIP ( DimCustomer[TerritoryKey],
                          DimTerritory[TerritoryKey] ) )
```

06 Time

```
Sales YTD =
    TOTALYTD ( [Total Sales], DimDate[Date] )

Sales QTD =
    TOTALQTD ( [Total Sales], DimDate[Date] )

Sales MTD =
    TOTALMTD ( [Total Sales], DimDate[Date] )

Sales FYTD =
    TOTALYTD ( [Total Sales], DimDate[Date], "31/03" )

Sales PY =
    CALCULATE ( [Total Sales],
                SAMEPERIODLASTYEAR ( DimDate[Date] ) )

Sales PM =
    CALCULATE ( [Total Sales],
                DATEADD ( DimDate[Date], -1, MONTH ) )

Sales PQ =
    CALCULATE ( [Total Sales],
                DATEADD ( DimDate[Date], -1, QUARTER ) )

Sales PY YTD =
    CALCULATE ( [Sales YTD],
                SAMEPERIODLASTYEAR ( DimDate[Date] ) )

YoY Growth =
    VAR Cur = [Total Sales]
    VAR Prev = [Sales PY]
    RETURN
        IF ( NOT ISBLANK ( Prev ), Cur - Prev )

YoY Growth % =
    DIVIDE ( [Total Sales] - [Sales PY], [Sales PY] )

MoM Growth % =
    DIVIDE ( [Total Sales] - [Sales PM], [Sales PM] )

YoY Trend =
    VAR G = [YoY Growth %]
    RETURN
        SWITCH ( TRUE (),
            ISBLANK ( G ), "No prior year",
            G > 0.10,      "Strong growth",
            G > 0,         "Growing",
            G > -0.10,     "Slight decline",
            "Sharp decline" )

Sales Rolling 3M =
```

```
    CALCULATE ( [Total Sales],
        DATESINPERIOD ( DimDate[Date], MAX ( DimDate[Date] ),
                        -3, MONTH ) )

Sales Rolling 12M =
    CALCULATE ( [Total Sales],
        DATESINPERIOD ( DimDate[Date], MAX ( DimDate[Date] ),
                        -12, MONTH ) )

Sales Cumulative =
    CALCULATE ( [Total Sales],
        FILTER ( ALLSELECTED ( DimDate[Date] ),
                 DimDate[Date] <= MAX ( DimDate[Date] ) ) )

Sales 3M Moving Avg =
    AVERAGEX (
        DATESINPERIOD ( DimDate[Date], MAX ( DimDate[Date] ),
                        -3, MONTH ),
        CALCULATE ( [Total Sales] ) )

Category Revenue PY =
    CALCULATE ( [Total Sales],
        ALLEXCEPT ( DimProduct, DimProduct[Category] ),
        SAMEPERIODLASTYEAR ( DimDate[Date] ) )

Electronics YoY % =
    VAR Cur =
        CALCULATE ( [Total Sales],
                    DimProduct[Category] = "Electronics" )
    VAR Prev =
        CALCULATE ( [Total Sales],
                    DimProduct[Category] = "Electronics",
                    SAMEPERIODLASTYEAR ( DimDate[Date] ) )
    RETURN
        DIVIDE ( Cur - Prev, Prev )

Best Month Sales =
    MAXX ( VALUES ( DimDate[YearMonth] ),
           CALCULATE ( [Total Sales] ) )

Best Month Name =
    VAR T = TOPN ( 1, VALUES ( DimDate[YearMonth] ),
                   CALCULATE ( [Total Sales] ), DESC )
    RETURN
        CONCATENATEX ( T, DimDate[YearMonth], ", " )
```

Calculated tables

Product Category Summary

```
Product Category Summary =
    SUMMARIZE (
        FactSales,
        DimProduct[Category],
        "Total Sales",         [Total Sales],
        "Total Quantity",      [Total Quantity],
        "Order Count",         [Order Count],
        "Avg Sales per Order", [Avg Sales per Order]
    )
```

Top 5 Products

```
Top 5 Products =
    TOPN (
        5,
        SUMMARIZE (
            FactSales,
            DimProduct[ProductName],
            DimProduct[Category],
            "Units Sold", [Total Quantity],
            "Revenue",    [Total Sales]
        ),
        [Units Sold], DESC
    )
```

Relationships

Name	From	To	Active
`rel_Date_Fact`	`FactSales.DateKey`	`DimDate.DateKey`	Yes
`rel_Product_Fact`	`FactSales.ProductKey`	`DimProduct.ProductKey`	Yes
`rel_Customer_Fact`	`FactSales.CustomerKey`	`DimCustomer.CustomerKey`	Yes
`rel_Territory_Fact`	`FactSales.TerritoryKey`	`DimTerritory.TerritoryKey`	Yes
`rel_Territory_Customer`	`DimCustomer.TerritoryKey`	`DimTerritory.TerritoryKey`	No

Figures and screenshots. Figures 1 to 4 are rendered directly from the actual dataset and are therefore genuine results. Figure 5 is an illustration of the report layout. The 3 screenshots are Power BI Desktop captures from the working file. The DAX appendix was extracted from the applied semantic model, so it is verifiably the code running in the report.